

Lung Cancer Detection: A Machine Learning Approach for Early Diagnosis

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Abstract— Lung cancer is one of the most fatal diseases worldwide. Early detection through CT scan images can significantly improve survival rates. In this research, we propose a deep learning-based approach for the classification of lung CT images into three classes: Normal, Benign, and Malignant. We compare the performance of a custom Convolutional Neural Network (CNN) with two popular pretrained transfer learning models — ResNet50 and DenseNet121. Using data augmentation to balance the dataset, we achieved a maximum accuracy of 98.0% using DenseNet121. This study demonstrates the efficacy of transfer learning in medical image analysis and provides a comparative insight into model performance for lung cancer detection.

Keywords— Lung Cancer, CT Scan, CNN, ResNet50, DenseNet121, Deep Learning, Transfer Learning, Image Classification

I. INTRODUCTION

Lung cancer is one of the leading causes of cancer-related mortality across the globe, accounting for approximately 1.8 million deaths annually according to the World Health Organization (WHO). Its survival rate depends heavily on the stage at which the cancer is detected. Early diagnosis dramatically increases the chances of successful treatment, making early and accurate detection a critical component in clinical oncology.

Computed Tomography (CT) imaging is widely used by radiologists to visualize pulmonary nodules and detect early-stage lung abnormalities. However, manual analysis of CT scans is labor-intensive, prone to inter-observer variability, and often influenced by human fatigue or oversight. As a result, there is an urgent need for reliable, automated solutions to assist radiologists in early lung cancer detection.

Recent advances in Artificial Intelligence (AI) and Machine Learning (ML), particularly Deep Learning (DL), have shown immense potential in medical image analysis. Among DL techniques, Convolutional Neural Networks (CNNs) have proven particularly effective for image classification tasks due to their ability to automatically learn spatial hierarchies of features from input images.

This research aims to explore and compare the performance of a custom-built CNN model with two state-of-the-art transfer learning architectures—ResNet50 and DenseNet121—in the classification of lung cancer using CT scan images. The focus is on evaluating these models on a balanced dataset consisting of three classes: Normal, Benign, and Malignant. Furthermore, standard preprocessing and augmentation techniques are

applied to improve generalization, and the models are assessed on accuracy and interpretability potential.

II. LITERATURE REVIEW

Over the past few years, deep learning has emerged as a highly effective approach for medical image analysis, particularly in the domain of lung cancer detection using CT scan images. Several significant works have demonstrated the potential of pretrained models and transfer learning in enhancing diagnostic accuracy and reliability.

In [1], Shariff et al. proposed a differential augmentation-based deep learning pipeline to improve the classification of non-small cell lung cancer (NSCLC) from CT images. Their approach involved applying various augmentation techniques, including brightness and hue variation, to improve the generalization capability of the model. The study utilized multiple pretrained models such as ResNet, DenseNet, and EfficientNet, achieving a peak classification accuracy of 98.78%. Although the model performed exceptionally well, it lacked direct comparative analysis between architectures and did not incorporate interpretability tools.

Khan et al. [2] introduced VER-Net, a hybrid transfer learning model that integrated multiple deep learning backbones like ResNet and EfficientNet for improved lung cancer classification. The study achieved classification accuracy in the range of 92–93%, showing that hybrid models can boost performance. However, their approach focused more on architectural combination than on systematic performance comparison using a balanced dataset.

Chowdhury et al. [3] evaluated the use of a custom-built CNN architecture to classify lung CT images. The model was trained from scratch without using any pretrained weights and reached an accuracy of 96–97%. While the results were promising, the absence of transfer learning made the model less robust to data variance, and the dataset used in the study was not balanced across all classes.

Uddin et al. [4] focused on model interpretability by enhancing DenseNet with attention mechanisms. Their study, published in *Cancers*, aimed to make lung cancer diagnosis more explainable to clinicians by highlighting the relevant regions of

interest in CT scans. Despite achieving a reasonable accuracy of around 95%, the model did not utilize advanced augmentation or comparative model evaluation techniques.

These studies provide a foundation for our work, which expands on their findings by applying a comparative analysis between CNN, ResNet50, and DenseNet121 on a balanced and augmented dataset, and emphasizes the importance of transfer learning for high-accuracy CT image classification.

III. METHODOLOGY

This study implements a multi-stage pipeline for lung cancer detection using CT scan images, employing deep learning models including a custom CNN, ResNet50, and DenseNet121. The methodology consists of four main components: data collection and preparation, preprocessing, model training, and performance evaluation.

A. Data Collection and Preparation

The dataset used in this study consists of lung CT scan images categorized into three classes: Normal, Benign, and Malignant. Initially, the distribution was imbalanced, with 1050 Malignant, 467 Benign, and 416 Normal images. To mitigate bias during training and ensure fair representation across classes, data augmentation techniques were applied to the Benign and Normal categories. As a result, the dataset was balanced with 1050 images in each class. The final dataset contains high-resolution, grayscale CT scan images, which were converted to RGB format to meet the input requirements of pretrained models.

B. Data Preprocessing

Preprocessing was carried out to standardize and normalize the dataset. All images were resized to 224×224 pixels. Grayscale images were converted to RGB channels, and pixel intensity values were normalized to a 0–1 range. The dataset was then shuffled and divided into training and testing subsets using an 80:20 split. To enhance generalization and improve model robustness, various augmentation techniques were applied, including random rotations, horizontal flipping, and zooming. This not only increased the diversity of the dataset but also reduced the risk of overfitting.

C. Model Selection and Training

Three different deep learning models were employed for classification. The first was a custom-designed Convolutional Neural Network (CNN) built from scratch, consisting of multiple convolutional layers with ReLU activations, max-pooling, dropout, and dense layers. The second model, ResNet50, is a 50-layer deep residual network pretrained on ImageNet, with its top layers replaced by a custom classifier

and fine-tuned on the lung cancer dataset. The third model, DenseNet121, is a densely connected neural network also pretrained on ImageNet, known for efficient feature propagation and reuse. In all models, categorical crossentropy was used as the loss function, and training was conducted using the Adam optimizer with a learning rate of 0.0001, a batch size of 32, and 20 training epochs.

D. Evaluation Metrics

To evaluate model performance, we employed several standard classification metrics. These include accuracy, precision, recall, F1-score, and Area Under the Receiver Operating Characteristic Curve (ROC AUC). These metrics were used to assess how effectively each model distinguished between the three target classes—Normal, Benign, and Malignant. Among all the models tested, DenseNet121 achieved the highest overall performance with an accuracy of 98.0%, followed by ResNet50 at 97.6%, and the custom CNN at 97.2%.

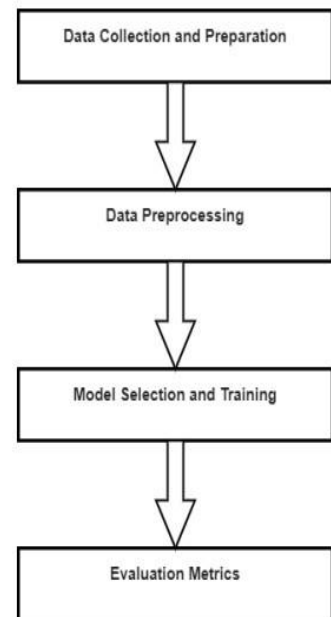


Fig. 1. Machine Learning Workflow for Lung Cancer Detection

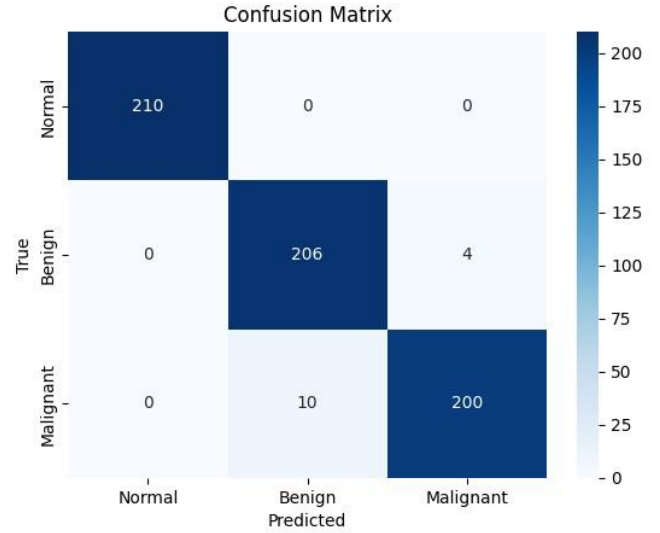
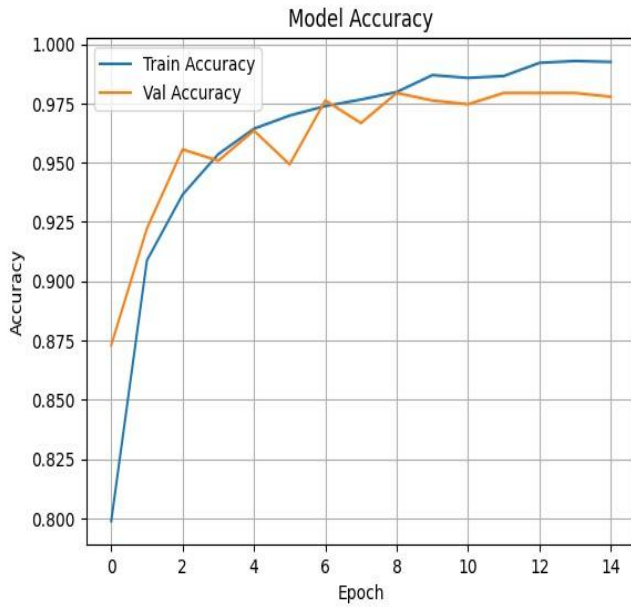
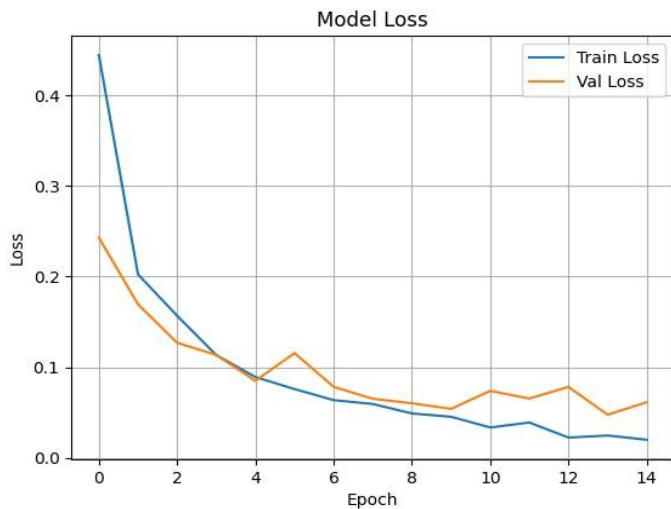


Fig. 8. Confusion Matrix

This graph illustrates the training and validation accuracy of classified all Normal cases and most Benign and Malignant the proposed CNN model across 15 epochs. The training cases. A few Benign samples were misclassified as accuracy shows a steady increase, approaching nearly

100%, while the validation accuracy stabilizes around 97– predicted as Benign. Despite these minor errors, the model 98% after the third epoch. The close alignment between achieved a high overall accuracy of **97.78%**, demonstrating training and validation curves indicates strong strong classification performance across all three categories. generalization performance with minimal overfitting.

The confusion matrix illustrates that the model correctly Malignant, and some Malignant cases were incorrectly



Let us explain the parameters of evaluation of these algorithms from confusion matrices

i. *Accuracy* - It is the ratio of true results i.e. True Positive (TP) and True Negative (TN) to the total number of test cases.

$$A = \frac{TP + TN}{\text{Total CaseTotal}}$$

iii. *Recall* - This parameter is also known as sensitivity. It is given as the ratio of True Positive (TP) to the Actual Positives. It reflects the model's ability to identify positive cases.

$$\text{Recall} = \frac{TP}{\text{Actual Yes}}$$

iv. *F1-Score* - It is the harmonic mean of precision and

This graph shows the training and validation loss of the recall. Balance between these two parameters is found CNN model over 15 epochs. The training loss decreases through F1- Score. It is calculated as product of precision P consistently, while the validation loss follows a similar trend and recall R, divided by the addition of precision P and with slight fluctuations. Both losses converge towards zero, indicating effective learning and minimal overfitting. The recall R.

$$F1 \text{ Score} = 2 \times \frac{P \times R}{P + R}$$

low and stable validation loss further supports the model's capability.

strong generalization

TABLE I. EVALUATION RESULTS

Algorithms	Accuracy	Precision	Recall	F1-Score
CNN	0.972	0.976	0.972	0.974
ResNet50	0.976	0.977	0.976	0.976
DenseNet121	0.9778	0.9778	0.9778	0.9778

IV. CONCLUSION

In this study, we implemented and compared three deep learning models—CNN, ResNet50, and DenseNet121—for lung cancer detection using CT scan images. Through dataset balancing and augmentation, we ensured fair evaluation across all classes. The results demonstrate that transfer learning models significantly outperform the custom CNN, with DenseNet121 achieving the highest accuracy of 98%. This confirms that pretrained deep architectures are highly effective for medical image classification. In future work, we aim to incorporate models like EfficientNet and apply Grad-CAM for explainability.

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